GAUSSIAN ELIMINATION ON ONE PAGE

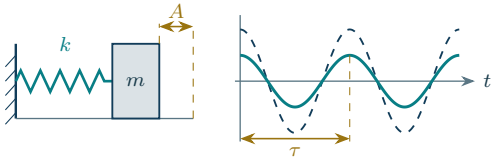
The period of a stiff spring

Dimensional analysis as a linear system, solved with every pivot and row swap in view, delivering $\tau \propto \sqrt{m/k}$.

1 / THE RULE AND THE PICTURE

Units become a matrix, and its solutions become formulas

A product $\Pi = q_1^{x_1} \cdots q_n^{x_n}$ has no units exactly when the exponents of mass, length and time all cancel, which is the linear system $\boxed{A\mathbf{x} = \mathbf{0}}$ whose column j holds the unit vector of q_j, and Buckingham's theorem counts its solutions: $n - \operatorname{rank} A$ independent dimensionless groups.



dashed: larger A
solid: smaller A
same τ

quantity	defining equation	units	$(\mathrm{M}, \mathrm{L}, \mathrm{T})$
m mass	base unit	M	$(1, 0, 0)$
k stiffness	$F = kx$	$(\mathrm{MLT}^{-2})/\mathrm{L}$	$(1, 0, -2)$
τ period	time to repeat	T	$(0, 0, 1)$
A amplitude	largest x	L	$(0, 1, 0)$

2 / THE CLASSICAL ROUTE

Equate exponents, then substitute

Try $\tau = C\, m^a k^b A^c$ with C a pure number. The right side has units $\mathrm{M}^{a+b}\mathrm{L}^{c}\mathrm{T}^{-2b}$, which must equal T^1, so

$$\mathrm{M}: \ a + b = 0, \qquad \mathrm{L}: \ c = 0, \qquad \mathrm{T}: \ -2b = 1.$$

Substituting upwards from the last equation gives $b = -\frac{1}{2}$, $a = \frac{1}{2}$, $c = 0$; the order of attack is a judgement, which the matrix route turns into rules.

3 / A DICTIONARY FOR THE MATRIX ROUTE

Every row operation is what you already do

swap two rows	write the equations in a different order
scale a row	multiply an equation by a non-zero constant
add a multiple	subtract equations to cancel an unknown
pivot	the unknown that a row solves for
free column	an unknown left as a parameter

A pivot must be non-zero, so a 0 on the diagonal is cured by swapping a non-zero entry up from below, which is the whole of pivoting.

4 / THE MATRIX ROUTE WITH THE PIVOTS IN VIEW

Four moves reduce the matrix, and the row labels travel with their rows

Rows are listed $\mathrm{L}, \mathrm{M}, \mathrm{T}$ so that the first column forces a swap; established pivots are gold and entries needing attention are red.

Step 1 / swap $R_1 \leftrightarrow R_2$

	m	k	τ	A
L	**0**	0	0	1
M	1	1	0	0
T	0	−2	1	0

$\Rightarrow$

	m	k	τ	A
M	**1**	1	0	0
L	0	0	0	1
T	0	−2	1	0

The corner entry is 0, so the pivot must come from lower down the column, and the M row supplies a 1.

Step 2 / swap $R_2 \leftrightarrow R_3$

	m	k	τ	A
M	**1**	1	0	0
L	0	**0**	0	1
T	0	−2	1	0

$\Rightarrow$

	m	k	τ	A
M	**1**	1	0	0
T	0	**−2**	1	0
L	0	0	0	1

The second pivot position also holds 0 while the -2 beneath it does not, so those rows trade places.

Step 3 / scale $R_2 \leftarrow -\frac{1}{2}R_2$

	m	k	τ	A
M	**1**	1	0	0
T	0	**−2**	1	0
L	0	0	0	1

$\Rightarrow$

	m	k	τ	A
M	**1**	1	0	0
T	0	**1**	−0.5	0
L	0	0	0	1

Dividing the pivot row by its pivot makes the pivot 1 and keeps the later arithmetic simple.

Step 4 / clear above $R_1 \leftarrow R_1 - R_2$

	m	k	τ	A
M	**1**	**1**	0	0
T	0	**1**	−0.5	0
L	0	0	0	1

$\Rightarrow$

	m	**k**	τ	**A**
M	**1**	0	0.5	0
T	0	**1**	−0.5	0
L	0	0	0	**1**

Column τ has no non-zero entry left in row 3, so it never gets a pivot and is free, while column A already holds its 1.

5 / READ THE ANSWER AND COUNT WITH BUCKINGHAM

Each pivot row solves for its own unknown

The rows read $m + \frac{1}{2}\tau = 0$, $k - \frac{1}{2}\tau = 0$, $A = 0$, so setting the free exponent to the wanted quantity, $x_\tau = 1$, gives $\mathbf{x} = (-\frac{1}{2}, \frac{1}{2}, 1, 0)$. With $n = 4$ columns and 3 pivots, $n - r = 1$ free column gives one group, $\Pi = \tau\sqrt{k/m}$, which is Buckingham's count.

$$\tau = \Pi\sqrt{m/k} \quad \text{with } A \text{ absent and } \Pi \text{ a pure number}$$

6 / CHECK AGAINST THE PHYSICS

What the pure number turns out to be

Newton's law $m\ddot{x} = -kx$ gives $\tau = 2\pi\sqrt{m/k}$, so $\Pi = 2\pi \approx 6.28$; for 0.50 kg on $200\ \mathrm{N\,m^{-1}}$, $\sqrt{m/k} = 0.050$ s and $\tau = 0.314$ s. Quadrupling k halves the period, quadrupling m doubles it, and the zero exponent of A makes the period independent of amplitude, as the two waves showed.

ASIDE / HANG THE MASS AND ADD g

Adding $g = (0, 1, -2)$ keeps three pivots but frees both τ and g, so $n - r = 2$ groups appear: $\Pi_1 = \tau\sqrt{k/m}$ and $\Pi_2 = mg/(kA)$, the static sag set against the amplitude. Units cannot discard Π_2; physics does, since gravity only shifts the equilibrium.

ASIDE / PIVOTING IN PYTHON

`sympy.Matrix(A).rref()` returns the reduced matrix and the pivot columns `(0, 1, 3)`, namely m, k, A. `scipy.linalg.lu(A)` returns P, L, U with $A = PLU$, where P records the row swaps; software takes the largest entry as pivot to limit rounding error.

Convention. Rank is the number of pivots, and the groups are the null space of A, one basis vector for each free column.
Follow-up: the density of the universe from units alone, where five quantities of rank three leave two groups.

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